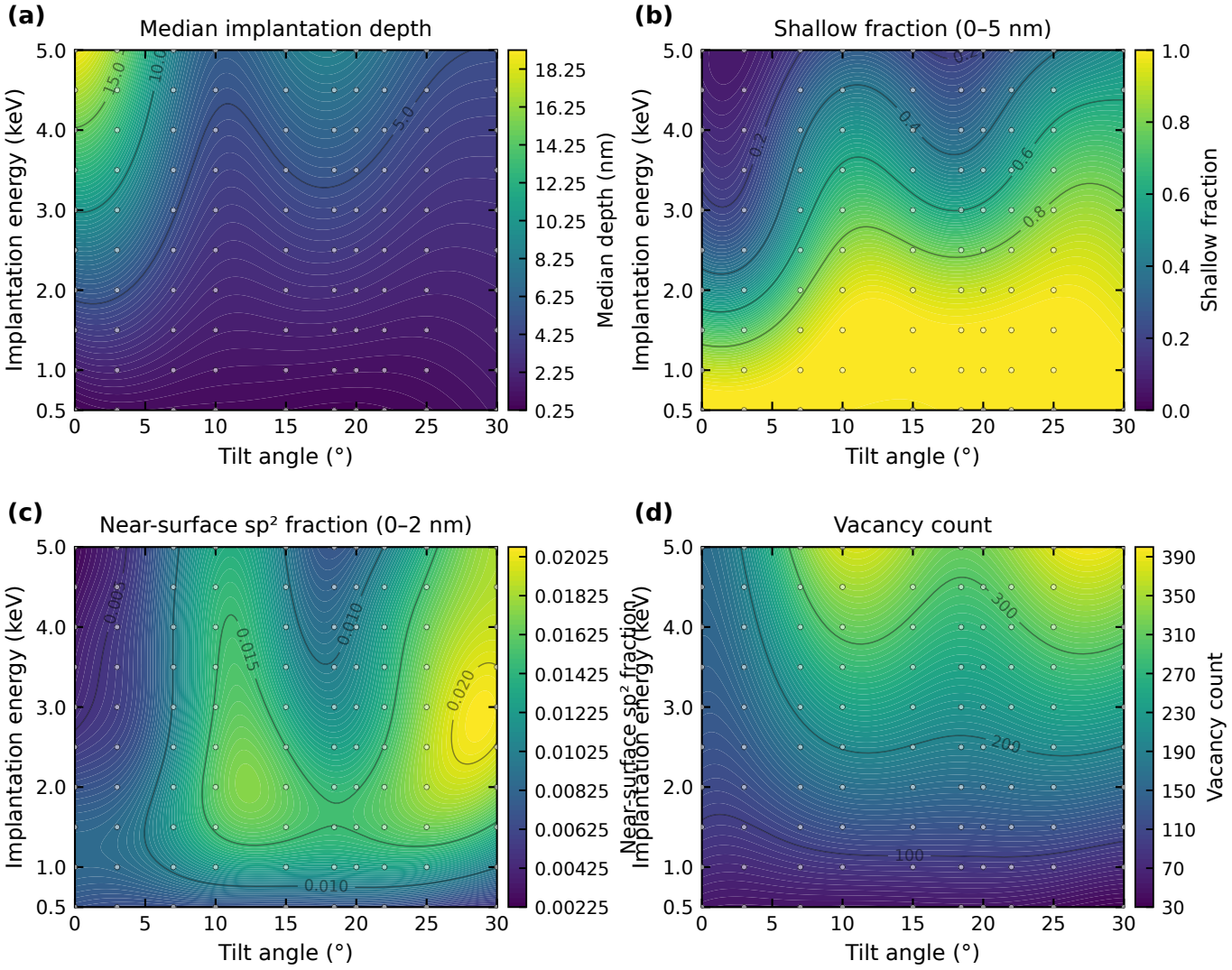

(a)
Median implantation depth
Implantation energy (keV)
Tilt angle (°)
Median depth (nm)
(b)
Shallow fraction (0–5 nm)
Implantation energy (keV)
Tilt angle (°)
Shallow fraction
(c)
Near-surface sp² fraction (0–2 nm)
Implantation energy (keV)
Tilt angle (°)
(d)
Vacancy count
Implantation energy (keV)
Tilt angle (°)
Vacancy count